

The Quantitative Soul: A Computational Comparison of Vietnamese Social Portraits under French Colonial Period

Mai Nong¹

¹Department of Network and Data Science, Central European University, Austria

Abstract

This study applies a computational sentiment analysis framework to examine the emotional landscapes of six seminal Vietnamese novels from the French colonial period. Utilizing the NRC Word-Emotion Association Lexicon and first-order Markov Chain modeling, the research quantifies how literary schools and social strata influenced narrative affect and structural rhythm. Findings reveal a clear emotional stratification: narratives of the oppressed peasantry and the impoverished intelligentsia are characterized by high-arousal negative emotions (fear and anger) and "stagnant" transition matrices, where emotional loops mirror material entrapment. In contrast, urban petty bourgeoisie and satirical narratives exhibit a more "liquid" emotional economy, defined by higher frequencies of trust and joy, and a more volatile narrative tempo. While a baseline of sadness was found to be a statistically universal constant across the corpus ($p = .0508$), the capacity for emotional transition was largely dependent on the protagonists' social strata. These results provide empirical evidence that colonial Vietnamese authors did not merely describe social realities but encoded the psychological boundaries of class and modernization into the very mathematical logic of their stories.

Introduction

The 1930-1945 period in Vietnam stands as a pivotal era of socio-political transition, marked by the deepening of French colonial structures and the radical modernization of the Vietnamese consciousness through the spread of the Romanized script [2]. This period also witnessed a divergence in literary philosophies as authors sought to document a society caught between decaying Confucian traditions and the encroaching Western modernity [4]. While traditional literary criticism has long categorized these authors into distinct schools, most notably the Romanticism of the *Tự Lực Văn Đoàn* and the Critical Realism of the Northern writers, the underlying emotional and thematic structures of their works remain largely quantified only through qualitative analyses [1].

This paper investigates the diverse literary approaches to portraying the Vietnamese colonial society through a computational lens, focusing on seminal works from six authors representing different literary schools: *Sống mòn* by Nam Cao [5], *Tắt đèn* by Ngô Tất Tố [6], *Ngày mới* by Thạch Lam [10], *Bước đường cùng* by Nguyễn Công Hoan [7], *Số đỏ* by Vũ Trọng Phụng [12], and *Đoạn tuyệt* by Nhất Linh [8]. These texts span the socio-economic spectrum, from the "impoverished

intelligentsia" (*trí thức nghèo*) and the "urban petty bourgeoisie" (*tiểu tư sản thành thị*), to the "oppressed peasantry" (*nông dân*) and the "urban underclass" (*tầng lớp dưới đáy thành thị*). By employing sentiment analysis techniques, this study aims to demonstrate that the "emotional signature" of a narrative is inextricably linked to the social class of its protagonist and the literature style influencing its author's writings. Ultimately, this research argues that while the colonial condition was a shared reality, the literary representation of that reality was fractured by class-based emotional trajectories and shaped under the divergent aesthetic and ideological lenses of the authors, illustrating how the intersection of literary movement, whether realism, romanticism, or satire, and the protagonists' socio-economic status fundamentally act as pieces to the collective picture portraying the colonial experience.

Background

Vietnamese Socio-Political Climate during 1930-1945

The period between 1930 and 1945 in Vietnam was defined by a profound "colonial paradox": the simultaneous advancement of Western modernization and the intensification of colonial exploitation [2]. Under the French administration, the traditional Confucian educational and administrative systems were dismantled, replaced by a bureaucracy that prioritized French linguistic and cultural hegemony. This shift was accelerated by the rise of the Romanized script (*Chữ Quốc ngữ*), which, while originally was a tool for colonial record-keeping and missionary work, was reclaimed by Vietnamese elites as a potent medium for mass literacy and the dissemination of modern ideals [4].

However, this modernization was starkly uneven. While the urban petty bourgeoisie and a new class of civil servants emerged in growing metropolitan centers like Hanoi and Saigon, the rural hinterlands remained shackled to a draconian tax systems and feudal land-tenure arrangements that had been exacerbated by colonial demands [13]. The 1930s specifically marked a decade of "tradition on trial", where the failure of early anti-colonial uprisings led to a period of deep introspection among the intelligentsia [4]. This group found themselves in a precarious position—educated in Western thought yet marginalized by the colonial hierarchy—leading to the "spiritual erosion" often depicted in the era's realist prose. Consequently, the social fabric was fragmented into distinct classes: a struggling peasantry, a stagnant impoverished intelligentsia, a reform-minded urban petty bourgeoisie, and a volatile urban underclass, each experiencing the pressures of the French rule through vastly different realities.

The Literary Landscapes of the Pre-1945 Era

The diversity of narrative voices in this period was not merely aesthetic but rooted in the varied backgrounds and lived experiences of the authors. Their literary styles, ranging from poetic introspection to biting satire—functioned as aesthetic responses to the material conditions of their respective social classes.

The Reformist Romantics: Individualism and Thesis Fiction

The Self-Reliance Literary Group (*Tự Lực Văn Đoàn*), represented by Nhất Linh and Thạch Lam,

emerged from the urban petty bourgeoisie. Nhất Linh, having received a formal French education in science and journalism, developed a "Thesis Fiction" (*luận đề*) style. This approach was characterized by a purposeful, reformist drive to dismantle feudal family structures in favor of individual autonomy [1]. In works like *Đoạn tuyệt* and *Lạnh lùng*, he purposefully staged conflicts between individual desire and stifling feudal traditions. His background as a political organizer and intellectual leader gave his prose a directional, optimistic reformist energy.

Conversely, Thạch Lam, though sharing this urban background, was more attuned to the psychological nuances of the impoverished intelligentsia. Influenced by French impressionism, his "Poetic Realism" eschewed overt plot in favor of capturing the subtle, fleeting emotions of life in the margins, as seen in his collections *Gió lạnh đầu mùa* and novel *Ngày mới*. For Thạch Lam, the style was a vehicle for empathy, capturing the quiet dignity found within the margins of poverty.

The Critical Realists: Documentary Truth and Spiritual Erosion

The Northern Realist school, including Ngô Tất Tố, Nguyễn Công Hoan, and Nam Cao, among others, was shaped by authors who lived in direct contact with the rural and suburban destitute. Ngô Tất Tố, a former Confucian scholar turned investigative journalist, utilized a "Documentary Realism" style in exemplary works like *Tắt đèn* and *Lều chõng*, to expose the crushing brutality of the colonial tax system on the oppressed peasantry and offer a sharp look at the decay of the Confucian examination system, respectively. This duality—the journalist's eye for current suffering and the scholar's critique of past structures—defined his signature style. Meanwhile, Nguyễn Công Hoan specialized in "Confrontational Realism", using sharp, rhythmic prose in works like *Bước đường cùng* and *Lá ngọc cành vàng* to highlight the hopeless cycles of debt facing the peasantry as well as the hypocrisy and cruelty of the upper class.

Nam Cao stood as the preeminent voice of the impoverished intelligentsia. As a struggling teacher frequently facing unemployment, Nam Cao's "Psychological Realism" focused on spiritual and moral erosion. In *Sống mòn* and his other works like *Chí Phèo* or *Đời thừa*, he explored the agonizing decay of human dignity under economic pressure. His style is defined by internal monologues and philosophical despair, mirroring the personal stagnation of an intellectual class marginalized by the colonial machine.

The Satirist: Grotesque Realism and the Urban Mirror

Vũ Trọng Phụng occupied a unique space in the literature landscape as the master of "Grotesque Realism". His background as a street-level reporter in Hanoi provided an unfiltered view of the urban underclass. Unlike earnest reformers, Phụng's style was celebratory of the absurd. In *Số đỏ* and his investigative reportage like *Kỹ nghệ lấy Tây*, he used biting satire and linguistic pastiche to portray a society where traditional values were discarded for hollow mimicry of Western "civilization" [4]. His "King of News" (*ông vua phóng sự đất Bắc*) background is evident in the rapid-fire, sensationalist pacing that captures the volatile energy of the colonial city.

The Selected Corpus: Narrative Frameworks and Social Stratification

To capture the multifaceted nature of colonial society, this study analyzes six seminal works selected for their representative value of specific social strata and literary movements. Each work provides a unique "emotional trajectory" that reflects the material reality of its protagonist's class.

1. *Tắt đèn* (Ngô Tất Tố)

- Protagonist: Chị Dậu (female, oppressed peasantry)
- Plot summary: Set against the suffocating rural tax system, the novel follows Chị Dậu's desperate struggles to pay the "poll tax" for her family. The narrative reaches a harrowing climax as she is forced to sell her child and her dog to a local landlord, only to be pushed further into darkness.

2. *Bước đường cùng* (Nguyễn Công Hoan)

- Protagonist: Anh Pha (male, oppressed peasantry)
- Plot summary: This work tracks the systematic ruins of Anh Pha, a hardworking farmer driven to bankruptcy by the predatory lending of the landlord class and the corrupt colonial legal system. The story captures the "dead end" (hence the name *Bước đường cùng*) of the rural poor, providing a baseline for themes of debt and structural exploitation.

3. *Sống mòn* (Nam Cao)

- Protagonist: Thứ (male, impoverished intelligentsia)
- Plot summary: Set in a cramped private school, the novel explores the stagnant life of Thứ, a teacher whose lofty ideals are slowly eroded by the trivial, selfish concerns of poverty. Unlike the high-stakes drama of the peasants, the conflicts here is internal, reflecting a spiritual and moral erosion where the protagonist feels himself rotting away in a meaningless existence.

4. *Ngày mới* (Thạch Lam)

- Protagonist: Trường (male, impoverished intelligentsia)
- Plot summary: Trường is an idealistic youth who rejects a wealthy arranged marriage to marry Trinh, a girl of modest means. However, the harsh "burden of rice and money" (*nỗi lo cơm áo gạo tiền*) soon transforms his gentleness into bitterness and resentment. As he and Trinh struggles to avoid the same cycle of misery that destroyed his brother's family, Trường is forced to confront the destructive power of poverty on the human soul. While the narrative dwells on this psychological erosion, it ultimately seeks a path toward spiritual renewal and a "new day" of inner resilience.

5. *Đoạn tuyệt* (Nhất Linh)

- Protagonist: Loan (female, urban petty bourgeoisie)
- Plot summary: Loan is an educated, modern woman forced into a traditional marriage with Thân, a wealthy but uneducated man, to secretly settle a family debt. Her domestic life becomes a nightmare of abuse by her husband and her traditionalist mother-in-law.

After suffering a miscarriage and further humiliation, Loan accidentally kills Thân during a violent confrontation. Her subsequent trial and acquittal symbolize a legal and social "severing" (*đoạn tuyệt*) of feudal ties, ending with Loan clearing her family's debts and reuniting with her true love, Dũng.

6. *Số đỏ* (Vũ Trọng Phụng)

- Protagonist: Xuân Tóc Đỏ (male, urban underclass)
- Plot summary: After being bailed out of jail for a petty offense, a street-urchin named Xuân is thrust into Hanoi's elite through the "Westernization" movement. By mimicking medical jargon and exploiting the vanity of the nouveau riche, he is rebranded as a "doctor" and a social reformer. His accidental "achievements" propel him to national heroism and a high-society marriage. The novel is a biting satire of colonial opportunism and the hollow, often ridiculous mimicry of Western "civilization."

Methodology

Data Acquisition and Corpus Preparation

The empirical foundation of this study consists of a specialized corpus of six canonical Vietnamese novels published between 1930 and 1945 introduced above. The primary texts were sourced from digitized literary archives and standardized into machine-readable plain-text (UTF-8) format. To facilitate a longitudinal analysis of the narrative sentiment, each work was systematically segmented following the original works' chapter systems using a special delimiter [CHAPTER_BREAK]. This structural preparation ensures that the subsequent natural language processing pipeline can analyze emotional trajectories as discrete temporal units, rather than as static aggregates.

Data Preprocessing for Vietnamese Text

The preprocessing of Vietnamese literary text presents unique challenges, primarily due to the language's monosyllabic script while meaning is frequently constructed through multi-syllabic compounds. To resolve this, the pipeline utilizes the `underthesea` library developed specifically for Vietnamese NLP [11] for sophisticated word segmentation and tokenization. A critical noise-reduction step involves the use of Named Entity Recognition to identify and automatically remove person names (tagged as B-PER or I-PER by `underthesea`, ensuring that subsequent LDA topic models focus on socio-economic and thematic vocabulary rather than specific character names.

Linguistic normalization is further achieved by replacing internal spaces in multi-syllabic tokens with underscores (e.g., *trí thức* becomes `trí_thức`) to preserve semantic units during analysis. Further more, to ensure the accuracy of sentiment assignment, a custom negation handling algorithm is applied: the particle *không* (no / not) is programmatically merged with its succeeding term (e.g., *không vui* becomes `không_vui`), preserving the intended valence for the lexicon-

based analyzer. Finally, the corpus is filtered using the `stopwordsiso` library [9] to remove the high-frequency functional words that do not contribute to thematic depth.

Lexicon Preparation

The sentiment analysis component of this research relies on VnEmoLex [3], a Vietnamese-specific adaptation of the NRC Emotion Lexicon. To ensure the lexicon’s compatability with the data processing pipeline, a dedicated preprocessing stage was conducted as follows: The raw dataset was cleaned to remove English translations and auxiliary metadata. The focus was restricted to the mapping of Vietnamese tokens to the eight primary emotions (anger, anticipation, disgust, fear, joy, sadness, surprise, and trust) as well as two overarching polarities, positive and negative. The processed tokens were converted into a custom JSON-based dictionary, in which each word acts as a key mapping to a nested dictionary of its associated emotional traits (e.g., `{ 'chua_xót' : { 'negative' : 1, 'disgust' : 1, 'sadness' : 1 } }`).

Sentiment Analysis Approach

Macroscopic Sentiment Analysis

To start with, the study performs a macroscopic analysis of each individual work to establish a baseline emotional profile. The raw frequency of emotion-bearing tokens is aggregated across the entire text of each novel. By mapping the preprocessed tokens of the work against the VnEmoLex dictionary, the study generates a cumulative sentiment score for the two overarching polarities and eight primary emotions. Based on the raw scores of the two polarities, the works’ overall sentiments are categorized into either "Positive" or "Negative" by comparing whether there are more positive tokens than negative tokens or vice versa.

However, this binary categorization does not quantify the degree to which a narrative leans toward a specific sentiment, the study employs a polarity measure as follows:

$$\text{Polarity} = \frac{\text{Total Positive Tokens} - \text{Total Negative Tokens}}{\text{Total Positive Tokens} + \text{Total Negative Tokens} + \varepsilon} \quad (1)$$

where ε is a small constant added to the denominator to prevent division by zero. The polarity measure yields a value between -1 and $+1$, where a negative value indicates a negative-dominant narrative and a positive value indicates a positive dominant narrative. This macroscopic metric serves as a quantitative "anchor," allowing for the objective comparison of dominant emotional atmospheres across different works featuring different literary styles and social classes.

Chapter-Level Sentiment Analysis

Following the baseline measurements across the entire work, the study also tracks the emotional evolution of each novel at a chapter level to capture narrative dynamics. Since chapter lengths can vary across the corpus, raw emotion-bearing token counts are normalized into emotional density by dividing by the chapter the chapter’s total token count. These sentiment density and polarity scores are then plotted sequentially to visualize the sentiment arcs of the novels. This longitudinal tracking provides a first step to explore how each author, within the context of their respective literary schools and protagonists, structure narrative development.

To further explore the emotional signature of each narrative and author, the study also compares the intensity distribution of each emotion across the novels with the Kruskal-Wallis H-test. This non-parametric method evaluates whether there is significant differences regarding a specific emotion among the six works. For the emotions yielding a significant result ($p < .05$), the Mann-Whitney U-test is utilized for pairwise follow-up comparisons between specific works. To account for the increased risk of Type I errors resulting from multiple testing, a Bonferroni correction is applied to the significance thresholds. These tests allows us to explore whether the emotional profile differ across different literary styles featuring different characters from different social strata.

While the emotional scores at the macroscopic level and chapter level reveal what emotions are present throughout the entire work and the chapter, respectively, they do not fully capture the narrative logic of how one emotion leads to another. Therefore, this study also employs first-order Markov Chains to quantify the "rhythm" and predictability of a narrative's emotional shifts. This approach treats each chapter's dominant emotion as a "state" and calculates the probability of transitioning from one state to another (e.g., the probability that a chapter dominant in fear is followed by a chapter dominant in sadness). The resulting emotional transition matrices provide a mathematical representation of the narrative structure, highlighting whether the narrative of interest is volatile, characterized by frequent transitions between disparate states, stagnant, defined by high self-transition probabilities or a consistent return to a specific "anchor" emotion.

Results

Macroscopic Sentiment Profiles

Novel	Positive	Negative	Sentiment Category	Polarity
Tắt đèn	677	753	Negative	-0.05
Bước đường cùng	1215	1311	Negative	-0.04
Sống mòn	1865	2145	Negative	-0.07
Ngày mới	1844	1409	Positive	0.13
Đoạn tuyệt	1191	1163	Positive	0.01
Số đỏ	1721	1388	Positive	0.11

Table 1: Macroscopic sentiment scores, polarity, and overarching sentiments

The raw global sentiment and polarity scores for each novel are reported in Table 1. Based on the raw counts of positive and negative tokens, the novels were categorized into two primary sentiment groups.

The macroscopic profiles reveal a clear bifurcation between the narratives of social stagnation and oppression versus those of modernization. While works depicting the "wear-and-tear" struggles, whether the internal psychological decay like in *Sống mòn* or outward struggles of the peasantry like in *Tắt đèn* and *Bước đường cùng*, yield negative-dominant sentiments, the reformist and satirical works focusing on Western modernization, such as *Đoạn tuyệt*, and *Số đỏ*, exhibit positive-

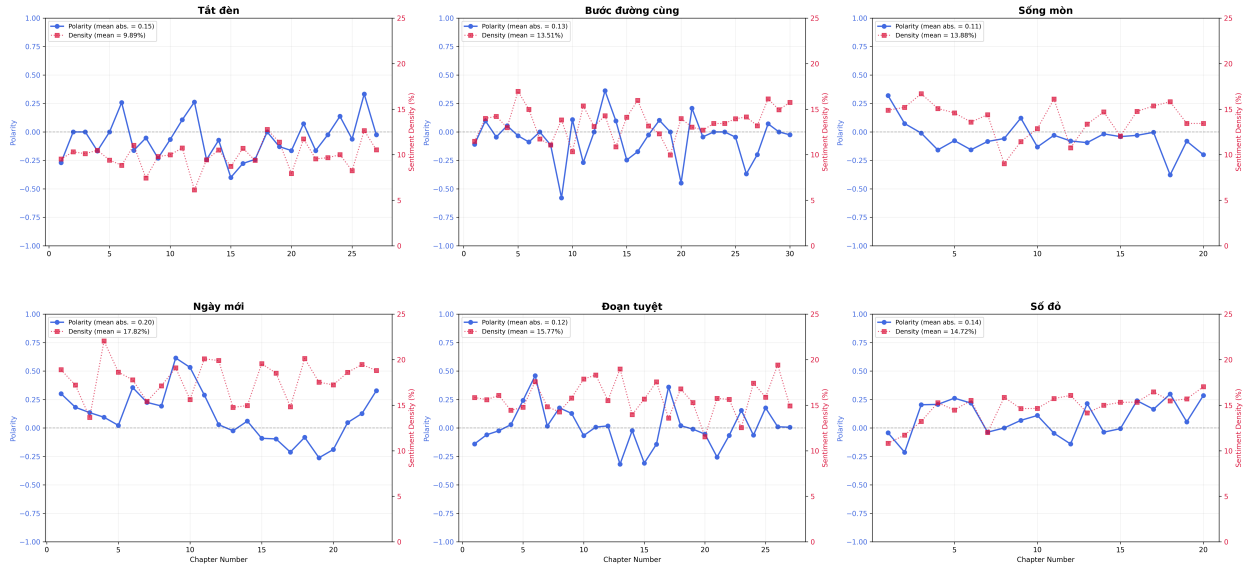


Figure 1: Chapter-level sentiment density and polarity trajectories for each novel, measured at chapter level. The left axis displays the polarity score, while the right axis corresponds to the normalized sentiment intensity.

dominant sentiments. Interestingly, *Ngày mới*, which explores the psychological erosion of an impoverished intelligentsia yet ultimately seeks spiritual renewal, stands out with the highest positive polarity score (0.13). This suggests that despite its focus on hardship, the narrative’s trajectory toward hope and resilience resonates strongly in its emotional profile, likely partially as a result of both the Tự Lực Văn Đoàn’s reformist school of thought and of Thạch Lam’s romantic idealism.

Narrative Sentiment Dynamics

The chapter-level sentiment density and polarity trajectories for each novel are visualized in Figure 1. Most works display relatively volatile polarity arcs fluctuating between positive and negative values. A notable exception is *Sống mòn*, with an almost flat polarity trajectory lingering just below 0 after the first few chapters, indicating a consistently negative-dominant narrative throughout. Meanwhile, for *Ngày mới*, the polarity arc initially rises steadily until chapter 10, before taking a sharp plunge over the subsequent two chapters. The trajectory then continues to decline at a slower rate, maintaining a negative sentiment towards the last few chapters, before starting to increase towards the positive territory again. This cyclical movement from idealistic optimism to profound crisis and back toward recovery not only mirrors the protagonist’s emotional journey but also reflects Tự Lực Văn Đoàn’s reformist agenda, where Thạch Lam’s romantic idealism ensures that the protagonist’s psychological descent acts as a transitional passage toward spiritual renewal rather than the terminal social stagnation characteristic of the realist school as seen in *Sống mòn*.

Unlike *Sống mòn*, the other two works on social stagnation and oppression, *Tắt đèn* and *Bước đường cùng*, exhibit much higher volatility in the polarity scores. Both narratives feature sharp peaks and deep troughs, with sentiments fluctuating at a much more rapid pace than the afore-

mentioned works. This high-frequency fluctuation suggests a narrative structure that oscillates between moments of despair and fleeting hope, reflecting the unpredictable and harsh realities faced by the peasantry under the colonial regime. This behavior also suggests that despite the characters' constant attempts to escape their dire circumstances, they are repeatedly confronted with further setbacks and challenges, only to remain systematically trapped in the same cycle.

On the other hand, the two works focusing on modernization, *Đoạn tuyệt* and *Số đỏ*, display a more moderated volatility in their polarity arcs. In the case of *Đoạn tuyệt*, the trajectory shows some fluctuation around the neutral line, except for the two notable peaks at chapters 6 and 17. *Số đỏ* also exhibits moderate volatility, but with the majority of its chapters leaning towards positive sentiment, though the majority of its chapters lean toward positive sentiment, a trend that persists even in the final chapter. This suggests that whether the author employs a reformist or satirical lens, narratives focusing on social change and modernization tend to maintain a linguistically optimistic tone.

Emotional Intensity Distributions

The results of the Kruskal-Wallis H-test (Table 2) indicate significant differences at 5% significance level in the distributions of most emotions across the six novels, except for anticipation and sadness, confirming that the emotional profiles of these works are distinct. The high variance, indicated by the effect size, in anger, fear, trust, and joy intensities suggests that these emotions are distributed unevenly across the corpus, likely reflecting the differing thematic focuses and narrative styles of the authors. Meanwhile, the non-significant results for anticipation and sadness indicates that these emotions are more uniformly represented across the works, possibly due to their universal relevance in the collective experience of Vietnamese society under the colonial regime.

Emotion	H-statistic	p-value	ϵ^2
Fear	23.4878	.000272	0.1311
Anger	35.4340	.000001	0.2158
Anticipation	6.0677	.299689	0.0076
Trust	28.5601	.000028	0.1671
Surprise	12.1945	.032218	0.0510
Sadness	11.0274	.050839	0.0427
Disgust	15.0316	.010228	0.0711
Joy	26.4575	.000073	0.1522

Table 2: Kruskal-Wallis H-test results and effect size (ϵ^2) for emotional intensity distributions across the six novels

The follow-up pairwise one-tailed Mann-Whitney U-tests with Bonferroni correction (Table 3) reveal specific inter-novel differences for the emotions that showed significant variance in the Kruskal-Wallis test. The result suggests that the emotional landscape of each novel is influenced by the social class of the protagonists and the literary schools of the writers.

Emotion	Novel A	Novel B	U-statistic	p-value	δ
Anger	Tắt đèn	Ngày mới	499.0	.000126	0.6071
Anger	Tắt đèn	Đoạn tuyệt	563.0	.000307	0.5446
Anger	Bước đường cùng	Ngày mới	575.0	.000019	0.6667
Anger	Bước đường cùng	Đoạn tuyệt	652.0	.000041	0.6099
Fear	Bước đường cùng	Sống mòn	462.0	.000691	0.5400
Fear	Bước đường cùng	Ngày mới	584.0	.000009	0.6928
Fear	Bước đường cùng	Đoạn tuyệt	598.0	.001047	0.4765
Trust	Ngày mới	Sống mòn	375.0	.000217	0.6304
Trust	Ngày mới	Bước đường cùng	576.0	.000018	0.6696
Trust	Số đỏ	Sống mòn	311.0	.001399	0.5550
Trust	Số đỏ	Bước đường cùng	491.0	.000081	0.6367
Trust	Đoạn tuyệt	Bước đường cùng	607.0	.000640	0.4988

Table 3: Significant results of the one-tailed Mann-Whitney U-tests for emotional intensity (Direction: $A > B$), including effect size (Cliff’s delta δ). Findings are filtered by a Bonferroni-corrected significance threshold of $\alpha = .0017$.

Anger emerges as a defining characteristic of the works of the featuring the oppressed peasantry, with *Tắt đèn* and *Bước đường cùng* exhibiting significantly higher anger intensities ($\delta > 0.54$) compared to works featuring the intelligentsia and the urban petty bourgeoisie. The high effect sizes also suggest that the anger expressed in works of the realist school is more pronounced and pervasive compared to those of the romantic reformists. Similarly, fear is also more intense in *Bước đường cùng* compared to works from other social strata, highlighting the acute, material vulnerability of the rural poor who continuously face the immediate threats of state and landlord exploitation.

In contrast, trust serves as the emotional signature of the works on modernization and social change. Works like *Ngày mới*, *Số đỏ*, and *Đoạn tuyệt* consistently exhibit higher trust intensities compared to those focusing on social stagnation and oppression. For the *Tự Lực Văn Đoàn* authors, this trust likely reflect their reformist optimism in the potentials for individuals to break free from feudal constraints and spiritual decay. In the case of Vũ Trọng Phụng, the elevated trust intensity in *Số đỏ* may be interpreted as a satirical commentary on the blind faith placed in Westernization and superficial social mobility, highlighting the absurdity of such aspirations in a colonial context.

Emotional Rhythms and Transitions Logic

Using first-order Markov Chains, the emotional transition matrices for each novel are calculated and visualized in Figure 2. By analyzing the self-transition probabilities (the diagonal) versus state-shift probabilities (the off-diagonal), the study identifies how social conditions dictate narrative movement.

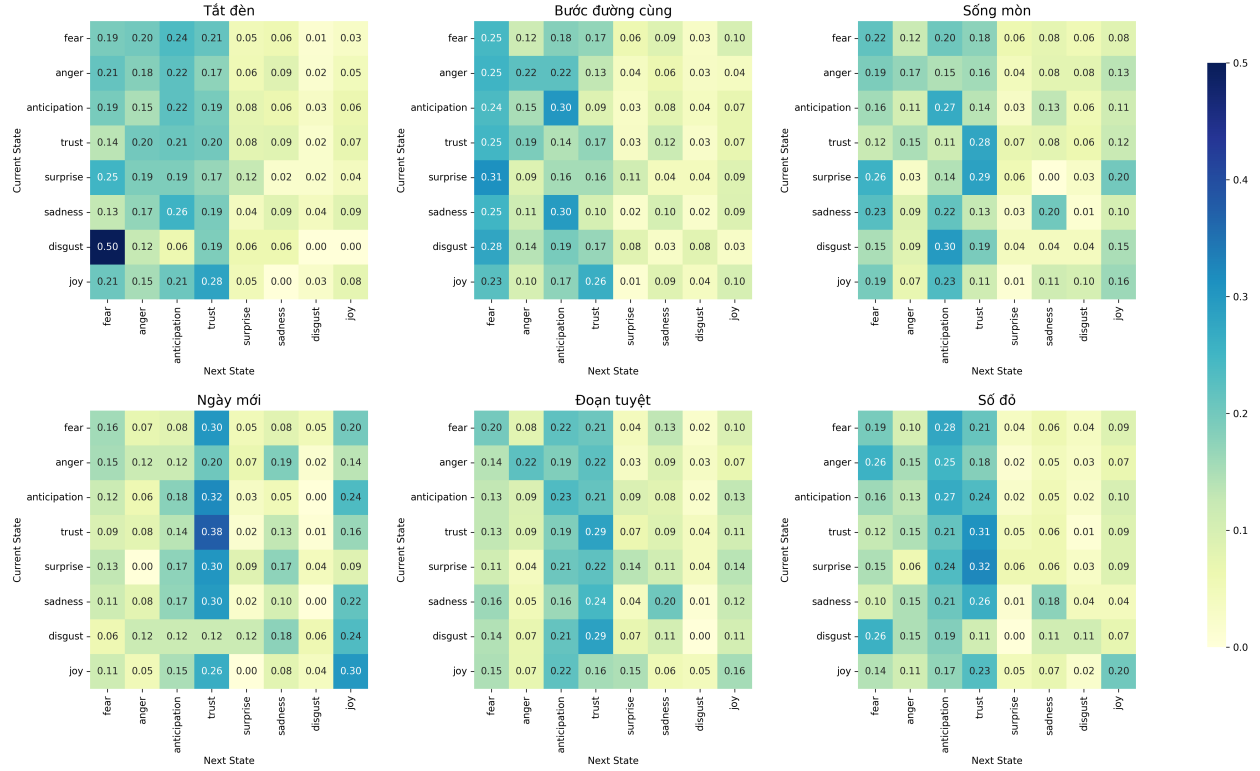


Figure 2: Emotional transition matrices for each novel, visualizing the probabilities of transitioning from one dominant emotion to another between chapters. Darker shades indicate higher transition probabilities.

For works depicting the oppressed peasantry, namely *Bước đường cùng* and *Tất đần*, the matrices reveal high self-transition probabilities among several negative emotions like fear and anger, as well as anticipation and trust. This suggests a narrative structure characterized by cycles of despair, anxiety, and fleeting hope, reflecting the relentless challenges faced by the rural poor. The frequent returns to anger and fear (especially in the case of *Bước đường cùng*) underscore a systemic entrapment within oppressive socio-economic conditions, where moments of optimism are quickly overshadowed by renewed hardship.

Sống mòn also exhibits high self-transition probabilities for negative emotions, particularly fear, sadness, and anticipation, which again, could be interpreted as feelings of anxiety and uncertainty. Interestingly, the narrative also shows a strong self-transition for sadness besides fear and anticipation, whereas the previous two works did not. This suggests that the psychological erosion experienced by the impoverished intelligentsia in Nam Cao's work is marked not only by fear and anxiety induced by external financial pressures but also by a deep-seated sadness stemming from internal moral decay and existensial despair. It also signals a more introspective narrative rhythm, where the protagonist's internal struggles are described as much as his external challenges.

In contrast to Nam Cao's approach, Thạch Lam's *Ngày mới* displays a more recurring pattern of emotional shifts, with trust and joy serving as emotional "anchors" that the narrative frequently returns to. This pattern more likely reflects the personal signature of Thạch Lam's romantic idealism,

where the presence of extreme negative emotions like fear, and anger are much less pronounced than joy and trust. The narrative rhythm here also suggest a more hopeful outlook, where despite the protagonist's psychological struggles, there is a persistent undercurrent of optimism and faith in the possibility of spiritual renewal.

Nhất Linh's *Đoạn tuyệt* while exhibits *Tự Lực Văn Đoàn*'s reformist optimism through high recurring probabilities for trust, also shows frequent recurring transitions into anticipation and high self-transition probabilities for fear, anger, and sadness. This pattern suggests that while the narrative is ultimately optimistic about human's triumph over feudal constraints, it does not shy away from depicting the emotional turmoil and societal resistance faced by the protagonist Loan as she navigates her oppressive marriage and traditional norms. The emotional rhythm here supports a complex interplay between hope for change and the harsh realities of entrenched social structures.

Finally, Vũ Trọng Phụng's *Số đỏ* presents a narrative rather similar to *Đoạn tuyệt*, with high recurring probabilities to trust and anticipation, reflecting a linguistically optimistic tone but also packed with suspense. However, unlike *Đoạn tuyệt*, it is uncertain to tell whether this linguistic optimism stems from genuine hope or satirical commentary without background knowledge of the work and its author. The narrative shows moderate self-transition probabilities across most emotions, but they are often outranked by other off-diagonal transitions, indicating a more dynamic and unpredictable emotional rhythm. This aligns with the satirical nature of the work, where the protagonist's rapid social ascension is marked by absurdity and chaos rather than steady progress.

Discussion

This study has utilized a computational sentiment analysis framework to map the emotional landscapes of seminal Vietnamese novels from the colonial period, revealing how literary styles and social strata shape narrative emotions. The oppressed peasantry narratives are defined by high-arousal negative affects (fear and anger) and structural stagnation, where emotional loops mirror material entrapment. The urban petty bourgeoisie and satirical works, on the other hand, exhibit more dynamic emotional rhythms with linguistic optimism (trust and joy). For the works on the impoverished intelligentsia, a dichotomy emerges: Nam Cao's *Sống mòn* reflects internalized despair and moral decay with a stagnant, linguistically negative tone, while Thạch Lam's *Ngài mới* offers a hopeful trajectory toward spiritual renewal. This suggests that whether a narrative operates within a "liquid" or "frozen" emotional economy depended not only on the literary school and the social class of the protagonist, but also on the author's personal philosophy and stylistic choices.

These findings largely align with existing qualitative assessments of the period's literature landscape, but also offer new quantitative evidence to support these interpretations. The distinct emotional signatures and rhythms identified through statistical testing and Markov Chain analysis provide a more nuanced understanding of how colonial Vietnamese authors navigated the complex interplay between social realities and literary expression. Furthermore, the study highlights the potential of computational methods in literary analysis, demonstrating how sentiment analysis can uncover latent emotional structures that traditional close reading might overlook. Future research could expand this approach to include a larger corpus and authors, and explore additional computational techniques, such as topic modeling or network analysis, to further elucidate the socio-cultural

dynamics of colonial Vietnamese literature.

However, it is important to acknowledge the limitations of this study. The reliance on a lexicon-based sentiment analysis approach may not fully capture the subtleties of literary language, particularly in a context as complex as colonial Vietnamese literature. Some nuances stemming from the use of French loanwords, which was very common in literature at the time, or regional dialects may not be fully captured by the preprocessing tools employed. The emotional connotations of words can vary significantly based on context, and certain literary devices, such as irony or metaphor, may not be adequately represented in the lexicon. The selection of novels, while representative, is limited in scope and may not encompass the full diversity of voices and styles present during the period, as the study does not include other formats such as poetry, music lyrics, or plays, as well as works of female authors, ethnic minorities, and Southern writers. Future studies could benefit from incorporating machine learning-based sentiment analysis techniques that can learn contextual nuances from the text itself, as well as expanding the corpus to include a broader range of authors and genres.

Conclusion

The computational analysis of these six seminal novels reveals that the emotional landscape of 1930s colonial Vietnam was not merely a backdrop for narrative, but a structured reflection of social stratification. Through the lens of sentiment analysis and Markov Chain modeling, this study has demonstrated that the oppressed peasantry and impoverished intelligentsia narratives were defined by emotional stagnation and negative feedback loops, while the urban petty bourgeoisie and underclass works operated within a more volatile, linguistically optimistic framework.

Ultimately, these findings prove that the colonial experience was marked by a fundamental divide in "emotional mobility." While a baseline of sadness remained a universal constant across all works, the ability to transition between emotional states, shifting from despair to hope or from fear to surprise, was a privilege largely reserved for the urban and reformist narratives. By quantifying these patterns, this study offers a new, empirical dimension to our understanding of how Vietnamese authors transformed the pressures of a changing society into the very structure of their stories.

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